Worksheet #19

Week 10, Day 1: Clustering

Name: _____
NetID: _____
Date: _____

Guided checkpoints

For full credit, fill in the checkpoints. If you skip a checkpoint, you must include your own notes for full credit.

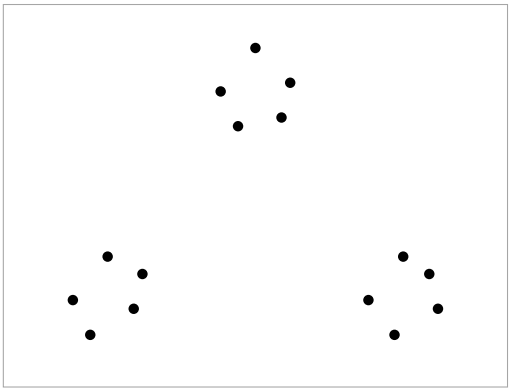
1. Supervised or Unsupervised?

Circle **S** (supervised) or **U** (unsupervised) for each task. If it is supervised, name the target variable.

Task	Type	Target (if supervised)
Predict a Cook County condo's sale price from its square footage and neighborhood.	S / U	
Group CTA stations whose weekday ridership looks alike.	S / U	
Predict a penguin's species from its bill and flipper measurements.	S / U	
Sort 4,500 movie reviews into groups without looking at their labels.	S / U	
Flag condo sales whose prices look unusual next to similar sales.	S / U	

2. Draw a Bad Start

Draw three starting centroids ($\times$) that would leave k-means ($K = 3$) stuck in a bad clustering. Then circle the three clusters it would end with.



3. Same Data, Different Answer

Two runs of k-means on the same penguins, both with $K = 3$, give different clusterings: Run A has SSE 206.1 and Run B has SSE 157.8. Which run would you keep, and why can the same algorithm give two different answers?

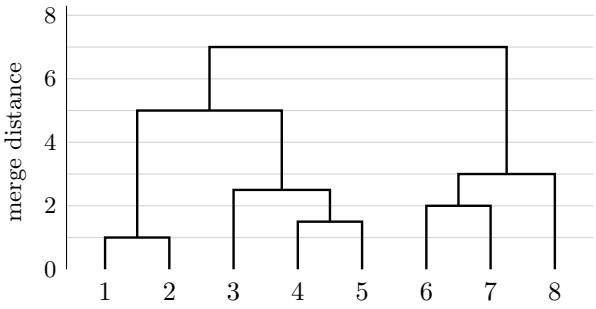
Name two ways to choose better initial centroids. What can go wrong with one of them?

4. Partitional or Hierarchical?

Circle **P** (partitional) or **H** (hierarchical) for the clustering each task needs, and say why in a few words.

Task	Type	Why?
Build a family tree of penguin species and their relatives.	P / H	
Split a delivery area into 5 driver zones.	P / H	
Organize songs into genres, subgenres, and sub-subgenres.	P / H	
Group customers into 3 segments for a mailing.	P / H	

5. Read a Dendrogram



Which two points are most similar? _____

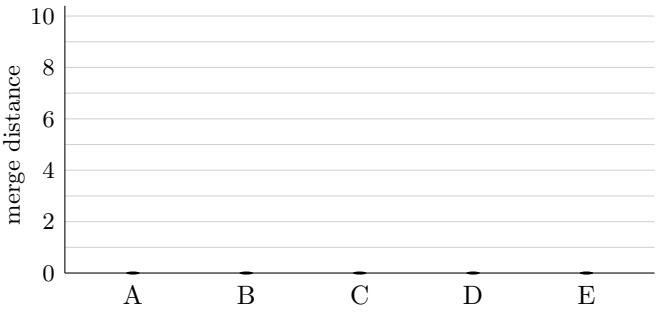
Draw a horizontal line that cuts the tree into 3 clusters. List them:

How many clusters does a cut at height 2.25 make? _____

6. Build a Dendrogram

Use **single link** (MIN) on the distances below. Draw each merge on the axes at the height where it happens. Then draw a horizontal line that cuts the tree into 2 clusters.

	A	B	C	D	E
A	0	2	6	10	9
B		0	5	9	8
C			0	4	7
D				0	3
E					0

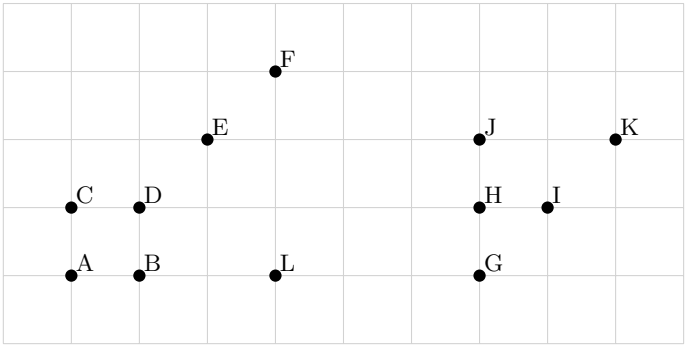


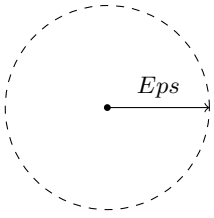
The 2 clusters are _____

With **complete link** (MAX) instead, which cluster does C join, and at what height?

7. Core, Border, Noise

Run DBSCAN with $Eps = 1.5$ grid squares and $MinPts = 4$, counting the point itself. Use the circle at right to check each neighborhood. Write **C** (core), **B** (border), or **N** (noise) next to each point. Then draw a loop around each cluster.





How many clusters did DBSCAN find? _____

F is only about 1.4 squares from E, and E is in a cluster. Why isn't F in that cluster?